

Tiny Object Detection

Architecture & Training Strategies

Deep Research Report



Faster R-CNN · RFLA Label Assignment · SAH-GD Metric · P2 FPN Level

Current best: HARD_SWITCH mAP(scale)=0.5770 | AP_micro=0.2776 | AP@75=0.0428 ← bottleneck

- #RoIAlign
- #CascadeRCNN
- #DioU
- #MultiScale
- #LabelAssignment
- #TinyObjects

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Project Context

Gaussian regression loss $1-\exp(-\beta \cdot D_H)$ is IoU-insensitive by design. This is the structural root cause of the stuck **AP@75 = 0.0428**. Dataset: 92% tiny/small, 27% micro (<8px), avg 45 obj/img. All recommendations in this report are anchored to this specific constraint.

01 High-Resolution RoIAlign for Tiny Objects

1.1 Why Standard 7×7 Fails for Micro Objects

Standard Faster R-CNN uses RoIAlign(output_size=7, sampling_ratio=2). For tiny objects, the number of feature cells covered by a proposal is far too small to carry useful spatial information.

Object Size	FPN Level	Stride	Feature Cells Covered	7×7 Pool Quality
32 px	P3	8	4×4	Moderate — upsampled from 4→7
16 px	P3	8	2×2	Poor — severe upsampling
8 px	P3	8	1×1	Useless — single cell → constant 7×7
8 px	P2	4	2×2	Still very coarse
6 px	P2	4	~1.5×1.5	Sub-pixel — only ~4 cells contribute

Key insight: Even at P2 (stride-4), a 6px object covers only ~1.5 feature cells. The 7×7 RoIAlign output is a bilinearly interpolated blow-up with almost no spatial variation — the box head has nothing useful to regress from. This directly explains the stuck AP@75.

1.2 Literature Evidence for Higher-Resolution RoIAlign

Paper	RoIAlign Config	Finding	Relevance
Mask R-CNN (2017)	14×14 RoIAlign	for mask head (keeps 7×7 for box head)	Baseline precedent
HRDNet (2021)	14×14 / 28×28	+1.5–2.0 AP on COCO small (14×14)	Direct evidence
Grid R-CNN (2019)	14×14 / 28×28 grid	+1.3 AP on COCO via grid-point prediction	Spatial exploitation
TridentNet (2019)	Scale-specific branches	Dilation-rate variant of the same principle	Scale-aware design

1.3 Per-Level Adaptive Strategy

The optimal output_size should vary by FPN level. The recommended approach uses 14×14 globally with a lightweight conv head (1–2 conv layers with stride 2) before the FC layers:

FPN Level	Stride	Typical Object	Feature Cells	Recommended output_size
P2	4	4–12 px	1–3 cells	14×14 or 21×21 — CRITICAL
P3	8	8–24 px	1–3 cells	14×14 — HIGH BENEFIT
P4	16	24–48 px	1.5–3 cells	7×7 — sufficient
P5	32	48–96 px	1.5–3 cells	7×7 — sufficient

1.4 Quantitative Impact Expectations

Configuration	Expected AP@75 Δ	Mechanism
7×7 → 14×14, plain FC head	+0.5–1.5 AP@75	More spatial info; FC may not exploit it
7×7 → 14×14, conv head (2–3 conv + pool)	+1.5–3.0 AP@75	Conv layers extract spatial patterns before regression
14×14 + sampling_ratio 2→4	+0.3–0.5 AP@75	Better sub-pixel interpolation quality
Per-level adaptive (14 for P2–P3, 7 for P4–P5)	Similar to global 14	Lower compute, avoids waste on easy levels

■ **Critical Caveat:** Higher RoIAlign alone is necessary but NOT sufficient. If the loss function does not reward precise localization, the head will not learn to use the additional spatial information. RoIAlign 14×14 MUST be combined with an IoU-aware regression loss (Section 03) to realize the AP@75 gain.

1.5 Verdict & Recommended Configuration

High Priority. The 7×7 RoIAlign is a clear structural bottleneck for objects <12px. Recommended implementation (Notebook 14 pattern):

Parameter	Value	Parameter	Value
RoIAlign output_size	14	sampling_ratio	4
Conv head	Conv3×3-256-ReLU → Conv3×3-256-ReLU → AdaptiveAvgPool(7)	fc	2×FC-1024
Must combine with	DIoU/CIoU regression term (Section 03)	Memory overhead	~4× per RoI (manageable)

02 Cascade R-CNN for Small/Tiny Objects

2.1 Core Mechanism

Cascade R-CNN (Cai & Vasconcelos, CVPR 2018) addresses the mismatch between training IoU thresholds and test-time quality. Each stage uses a progressively stricter IoU threshold, with proposals from the previous stage as input:

Standard	RPN → RoI Head (IoU=0.5) → output
2-stage Cascade	RPN → Stage 1 (IoU=0.5) → Stage 2 (IoU=0.6) → output
3-stage Cascade	RPN → Stage 1 (IoU=0.5) → Stage 2 (IoU=0.6) → Stage 3 (IoU=0.7) → output

2.2 The IoU Problem for Tiny Boxes

For tiny objects, IoU is extremely sensitive to pixel-level shifts. This fundamentally challenges the cascade assumption that successive stages can achieve higher IoU:

Object Size	Shift δ	Resulting IoU	Interpretation
32 px	1 px	0.88	Achievable with stride-8 features
16 px	1 px	0.77	Needs precision
16 px	2 px	0.59	1-stage threshold already hard
8 px	1 px	0.59	Sub-pixel precision required
8 px	2 px	0.36	Falls below even Stage 1 threshold
6 px	1 px	0.47	Already below IoU=0.5
6 px	2 px	0.24	Stage 3 at IoU=0.7 is impossible

2.3 Threshold Configurations

Config	IoU Thresholds	Pros	Cons
2-stage standard	[0.5, 0.6]	Conservative; positives at 0.6 for tiny	Still too tight for <8px
2-stage tiny-tuned	[0.4, 0.55]	More tiny positives at both stages	Recommended for this project
3-stage standard	[0.5, 0.6, 0.7]	Maximum refinement for large objects	Stage 3 starved of tiny positives
3-stage tiny-tuned	[0.4, 0.5, 0.6]	Reasonable tiny coverage	IoU=0.6 still challenging for <8px

2.4 Efficiency Considerations (Kaggle T4 constraint)

Component	2-Stage Overhead	3-Stage Overhead
Parameters	+50% RoI head	+100% RoI head
Training time	+30–40%	+50–70%
Inference time	+20–30%	+40–50%
Memory	+25–35%	+45–60%

2.5 Verdict

Medium priority — try after RoIAlign fix and regression loss fix. Cascade adds value but requires: (1) lowering IoU thresholds to [0.4, 0.55] for tiny scale, (2) adapting to Gaussian-distance matching if possible, (3) 2-stage preferred over 3-stage for memory and positive-sample reasons.

03 Regression Parametrization — THE AP@75 Lever

3.1 Root Cause Analysis

The project uses Gaussian similarity regression: $L_{reg} = 1 - \exp(-\beta \cdot D_H)$. This function has a fundamental structural property that prevents AP@75 from improving:

■ Scale-invariant	Good — no explosion for tiny boxes
■ Stable gradients	Good — no scale-dependent loss magnitude
■ IoU-insensitive	CRITICAL — a 1–2px error on a 6px box gives high similarity; loss is 'satisfied'
■ Vanishing gradient at extremes	Hard cases (highly misaligned proposals) get weak gradient signal

3.2 Comparative Loss Analysis for Tiny Box Regression

Loss	Scale Invariant?	Signal at Zero Overlap?	Precise at High IoU?	Stability <8px?
Smooth-L1 (standard)	No	N/A	Moderate	Poor
Gaussian similarity	Yes	Yes (Wasserstein)	NO — saturates	Excellent
GIoU	Yes	Weak (slow)	Good	Moderate
DIoU	Yes	STRONG (center)	Good	Good ✓
CIoU	Yes	Strong	Best (aspect ratio)	Moderate
NWD/GCD raw	Yes	Yes	No (same as Gaussian)	Excellent

3.3 The Dual-Objective Solution

The correct fix is to keep the Gaussian similarity term (for stability and scale-invariance) and add a **precision term** (IoU-aware) that activates at high overlap:

```
L_reg = (1 - S_H) # Gaussian similarity: coarse, stable, scale-invariant
+ γ · L_precision # IoU-aware precision term

# Recommended: DIoU loss
# DIoU = IoU - ρ²(b_pred, b_gt) / c²
# Center distance penalty provides gradient even at zero overlap
# Critical for 6–8px objects where initial proposals often have zero IoU
```

3.4 Why DIoU Over CIoU for This Project

For 6–8px objects, center penalty is the dominant need. A 2px center error → IoU of 0.24–0.36. CIoU adds an aspect-ratio term that is **quantization-sensitive for tiny boxes**: the difference between $\arctan(6/10)$ and $\arctan(7/10)$ is only 0.007 — dominated by pixel noise. DIoU's center penalty provides non-zero gradient for non-overlapping proposals, which is critical when initial RPN proposals are often completely misaligned with micro-object GTs.

3.5 Scheduling Strategy (Critical Implementation Detail)

Notebook 10 failure (GAMMA_FINE=1.0 from epoch 1) demonstrates that applying the IoU-aware term too early or too strongly causes training instability. Early proposals are so imprecise that IoU-based gradients are noisy. The recommended schedule:

```
if epoch < warmup_epochs (=3):
    L = 1 - S_H # Gaussian only
```

```
elif epoch < transition_end (=8):  
     $\gamma_{\text{eff}} = \gamma \cdot (\text{epoch} - 3) / (8 - 3)$  # linear ramp  
     $L = (1 - S_H) + \gamma_{\text{eff}} \cdot \text{DIoU\_loss}$   
else:  
     $L = (1 - S_H) + \gamma \cdot \text{DIoU\_loss}$  # full dual ( $\gamma = 0.3-0.5$ )
```

3.6 Verdict

HIGHEST PRIORITY — this is THE lever for AP@75. No architectural change can fix AP@75 if the loss doesn't reward precision. Recommended: DIoU as the precision term, $\gamma=0.3-0.5$, scheduled ramp-up starting epoch 3–4. Keep Gaussian similarity as the primary term for stability.

04 Multi-Scale Training & Inference Strategies

4.1 The Scale Problem

The TinyPerson dataset spans a 144× scale range (2px–289px). 27% of objects are micro (<8px), 52% are micro+tiny (<12px). Standard single-resolution training forces the network to handle all scales simultaneously — a structural disadvantage for micro-object recognition.

4.2 SNIP / SNIPER

Method	Core Idea	Assessment for This Project
SNIP (CVPR 2018)	Image pyramid — train only on objects in the 'valid range' for each scale	Good theory; prohibitive memory for TinyPerson (4–8× upscale needed)
SNIPER (NeurIPS 2018)	Chip-based SNIP — crop fixed-size chips around object clusters	More feasible; significant implementation complexity for RFLA + Faster R-CNN

4.3 Practical Multi-Scale Approaches

Strategy	Mechanism	Assessment	Complexity	Recommendation
Multi-resolution training (random scale)	Each iteration: random resolution from {512, 640, 768, 896, 1024, 1280, 1408, 1536, 1664, 1792, 1920} × X_SIZE already supported	Easy to implement; consistent improvement	Low	■ DO THIS
Copy-paste augmentation	Extract micro crops, paste at random positions	Maintains object size; more controlled than random crops	Medium	■ Worth trying
Mosaic (crop, not resize)	Combine 4 image crops without downsampling	Maintains pixel size; avoid resize variance	Medium	■ With care
Multi-scale TTA (inference)	Run at 1.0×, 1.5×, 2.0× and merge with NMS	Too slow for Kaggle time limit; useful for offline eval	High	■ Offline only

■ Mosaic warning: Standard mosaic resizes each quadrant, which for 1920×1080 images at 1024px target shrinks objects by ~2× (4px objects → 2px — undetectable). Always use the crop variant (512×512 patches).

4.4 Verdict

Medium priority, easy wins available. Multi-resolution training is the default that should have been enabled from the start — minimal code change, consistent small improvement. Copy-paste augmentation is the right choice for micro objects specifically. SNIP/SNIPER provide the largest theoretical benefit but require significant engineering effort — defer for now.

05 Label Assignment for Tiny Objects

5.1 Why Assignment Matters Disproportionately

For tiny objects, label assignment is critical for three compounding reasons:

#	Problem	Impact
1	Few anchors overlap	An 8px GT on P3 (stride-8) has IoU>0.5 with at most 1–2 anchors. Many GTs get ZERO positives under IoU-based assignment.
2	Extreme positive-negative ratio	160×90 feature map at P3 → 14,400 locations. 45 GTs/image → <0.3% positive ratio. Classifier learns 'everything is background'.
3	IoU is fundamentally unreliable	For a 6×6 box, 1px shift changes IoU from 0.69 to 0.56 — massive noise. IoU threshold assignment is dominated by quantization noise.

5.2 Method Comparison for <8px Objects

Method	Typical k (6px GT)	Quality Metric	Assigns 0 Positives?	IoU Dependence	Computation
Fixed IoU (0.5)	0–1	IoU	YES, often	Full	Low
ATSS (k=9)	3–5	IoU (adaptive thresh.)	Rare	High	Low
SimOTA	0–3	Cost (cls+reg)	YES if IoU=0	High	Medium
PAA	2–4	GMM on IoU×cls	Rare/noisy	High	High
Dynamic k (scale-adap.)	6–9	RF distance	NO ✓	None	Low
RFLA (this project)	3–9	Gaussian RF dist.	NO ✓	None ✓	Low

5.3 What Actually Works for <8px Objects

■ Replace IoU with scale-invariant metric (NWD/GCD) in matching.

Single most impactful change. IoU is fundamentally broken for <8px — quantization noise exceeds signal. ■ Already done in this project via RFLA + NWD/GCD.

■ Guarantee minimum positives per GT (k≥1).

SimOTA and PAA can fail to assign any positives for micro objects when all IoU < 0.1. Dynamic k with a floor is essential.

■ Scale-adaptive k (more positives for smaller objects).

Validated by project experiments. Micro objects need more positives: coarser features and noisier loss landscape.

SCALE_TOPK won AP_micro=0.2947 in SAH-GD ablation.

■ Anchor/feature resolution > assignment sophistication.

P2 result (+29% AP_micro) shows that giving micro objects actual feature resolution (stride-4 vs stride-8) has larger impact than any assignment change.

5.4 Recommended Configuration

```
# Label assignment: RFLA with scale-adaptive k
ASSIGNMENT_METRIC = 'gaussian_rf_distance' # NOT IoU
K_MICRO = 6 # objects < 8px (was 9; reduced to avoid noisy positives with P2)
K_TINY = 5 # objects 8–20px
K_OTHER = 3 # objects > 20px
MIN_K = 1 # guarantee at least 1 positive per GT
```

5.5 Verdict

Already well-addressed. RFLA + scale-adaptive k + Gaussian distance assignment is near-optimal for this dataset. Remaining gains are in: (1) tuning k values ($k_{\text{micro}}=6 > k_{\text{micro}}=9$ per notebook 12), (2) co-optimizing with P2 feature resolution, (3) potentially adding quality-weighted selection within top-k.

06 Synthesis & Priority Recommendations

6.1 Priority Matrix

Intervention	Expected AP@75 Δ	Expected AP_micro Δ	Complexity	Priority
DIoU regression (scheduled $\gamma=0.3-0.5$)	+2–4%	$\pm 0\%$	Low	■ HIGHEST
RoIAlign 14×14 + conv head	+1–2%	+1–3%	Medium	■ HIGH
DIoU + RoIAlign 14×14 combined	+3–5%	+1–3%	Medium	■ HIGH
Multi-resolution training	+0.5–1%	+1–2%	Low	■ MEDIUM
2-stage Cascade (tuned thresholds)	+1–3%	+0–1%	High	■ MEDIUM
Copy-paste augmentation	+0–1%	+2–4%	Medium	■ MEDIUM
Scale-adaptive k tuning	+0–0.5%	+1–2%	Low	■ LOW (already good)
3-stage Cascade	+1–2%	–1–0%	Very High	■ LOW
SNIP/SNIPER	+0.5–1%	+2–3%	Very High	■ LOW (defer)
Mosaic augmentation	+0–0.5%	+1–2%	Medium	■ LOW

6.2 Recommended Execution Order

Phase 1 (Fix AP@75 bottleneck)	1. (a) Diou regression with scheduled ramp-up ($\gamma=0.3$, warmup=3 epochs) — on clean P2F baseline 2. (b) RoIAlign 14×14 + conv head — on clean P2F baseline 3. (c) Combine winner of (a)+(b) → Expected: AP@75 > 0.06, mAP(scale) ≥ 0.58	1–2 experiments
Phase 2 (Stack micro gains)	4. (d) Scale-adaptive k tuning (k_micro=6) — on Phase 1 winner 5. (e) Multi-resolution training (random 640–1024) — on Phase 1 winner	2–3 experiments
Phase 3 (Advanced, only if plateau)	6. (f) 2-stage Cascade with Gaussian-distance thresholds 7. (g) Copy-paste augmentation for micro objects 8. (h) CIoU instead of Diou (if aspect ratio precision needed)	3–4 experiments

6.3 Key Principles

1. Fix the loss function first.

No architectural change can fix AP@75 if the loss doesn't reward precision. The Gaussian similarity loss is the root cause — adding a Diou term is the minimum necessary fix.

2. One variable at a time.

Notebook 9's P2 result was confounded by 5 simultaneous changes. Each intervention should be tested in isolation on the P2F baseline.

3. RoIAlign resolution and regression loss are complementary, not alternatives.

Higher RoIAlign gives the head more spatial information to work with; IoU-aware loss teaches it to use that information for precise boxes. Neither alone is sufficient.

4. Label assignment is already near-optimal.

RFLA + Gaussian distance + scale-adaptive k is well-suited to this dataset. Further assignment changes have diminishing returns — the bottleneck is now in the head.

5. Multi-scale training is easy wins.

Random resolution training requires minimal code changes and provides consistent small improvements. It should be the default, not an experiment.

References

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